

Roylan Martinez Vargas

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SUMMARY

Risk Data Specialist focused in the financial/banking sector, with expertise in agentic development, machine learning, data pipeline development and process mining. Interested in quantitative economics, with published research on payment clearing optimization cited by multiple universities. Actively engaged in independent research focusing on Ito calculus and central bank net compensation mechanisms.

EDUCATION

Mathematics, master's degree of research Valencia, Spain, 2025
Title jointly awarded by the University of Valencia and the Polytechnic University of Valencia

Relevant coursework: I investigated the Stochastic differential equations and how they affect mathematically the SOFR and EURIBOR (interest reference rates). My final thesis was a 9.5 out of 10.

Economics, bachelor's degree Barcelona, Spain, 2022
Title awarded by the Autonomous University of Barcelona

Relevant coursework: I investigated the optimization proposals to the payment clearing a work that I later have published and has been used by at **least three universities** (it can be verified in my google scholar profile: **Roylan Martinez**)

EXPERIENCE

Banco Sabadell, Barcelona, Spain Apr 2026-now
Risk Data Specialist– Model Risk Management

Supervise and govern the end-to-end lifecycle of all quantitative, analytical, and AI/cognitive models across the bank, including IRB, capital, IRRBB, traditional machine learning and agentic AI.

- Major accomplishments:
1. Actively shaping the strategic evolution of Model Risk Management processes in a highly regulated environment.

Allianz, Barcelona, Spain Nov 2025-Apr 2026
AI Data Scientist – Center of Excellence in Automation & AI

Develop and integrate AI/LLM capabilities into automated data pipelines to enhance operational efficiency and decision making across production deployments.

- Major accomplishments:
2. Engineered a high efficiency image processing workflow for LLM integration. Replaced proprietary cloud APIs with a raw byte stream chunking methodology, enhancing data throughput and eliminating dependency on high cost Azure specialized services for document OCR processing.

Allianz, Madrid, Spain April 2023-Feb 2024

Data Scientist – Quality and Fraud

- Involved in different projects related with fraud and financial crimes, quality and insurance recoveries. My responsibilities include data modelling, ETL pipeline development, cloud data analysis development and process mining.

Major accomplishments:

1. Developed an end-to-end machine learning solution to automate payment analysis and fraud detection, saving the Quality and Fraud department 1 FTE in manual labor.

PUBLICATIONS

Analysis of the Stochastic Modeling of EURIBOR and SOFR

Sep 2025

- The objective of this study is to examine the intrinsic differences associated with the calculation of EURIBOR and SOFR, exploring how these differences impact conventional modeling techniques --such as the deterministic approach--

Optimization Proposals to the Payment Clearing

Feb 2023

- The presented concepts--built upon solutions of relatively new but well-known graph theory and mathematical optimization theory--show how the use of some payment transaction methods can improve the traditional compensation logic behind a payment transaction.

Discrete Markov Chain Applications

March 2023

- The presented concepts--built upon solutions of relatively new but well-known graph theory and mathematical optimization theory--show how the use of some payment transaction methods can improve the traditional compensation logic behind a payment transaction. (This paper is not published within the associated journal yet)

SKILLS

Technical skills

- Programming Languages: Python (6+ years), SQL (especially Transact-SQL and PROC SQL), DAX, R, C.
- Cloud & Data Architecture: Microsoft Azure (Cloud Services, Synapse, Foundry), Docker, SAS, Celonis.
- Visualization & Reporting: Power BI.

Languages

- Spanish: Native
- English: Full Professional Proficiency
- Catalan: Full Professional Proficiency
- German: Basic